

# Beyond urban bias: A circulation-based model for equitable medical workforce distribution

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## ABSTRACT

**Purpose:** Urban–rural healthcare disparities remain a global challenge, driven by the concentration of medical resources in urban centers. We developed a circulation-based model integrating medical resource allocation, education, and digital health. Since 2020, this program has been implemented in Hokkaido, Japan. This study examines its structure and outcomes.

**Methods:** The model includes physician and trainee rotation, structured referral pathways, community-based medical education, and digital infrastructure. A three-tier system was established: Esashi (primary/secondary care), Hakodate (tertiary center), and Sapporo (education and physician dispatch hub). Outcomes were assessed in three domains: health outcomes, financial sustainability, and educational outcomes. Educational outcomes were based on a previously published study in the same cohort.

**Results:** Health outcomes showed overall improvement, with the standardized mortality ratio decreasing from 103.4 to 98.3 in males and from 104.8 to 100.5 in females, although these changes were not statistically significant. Pneumonia-related SMR decreased significantly in both males and females. Financial indicators improved, with inpatient revenue per patient increasing significantly from 26,553 to 38,478 JPY. In addition, medical students reported increased confidence in clinical competencies.

**Conclusion:** This model improved health, financial, and educational outcomes by coordinating physician mobility, tiered regional collaboration, and digital connectivity.

(195 words)

## 1. Introduction

Urban–rural healthcare disparities, rooted in urban-centric resource allocation, remain a global challenge. Physician density, training opportunities, and specialized medical services are typically concentrated in urban centers, leaving rural populations with reduced access to care.

In Japan, although universal health coverage has been achieved since 1961, medical workforce maldistribution remains a major structural challenge. Japan is also one of the most rapidly aging societies in the world, with more than 29% of the population aged 65 years or older. This demographic imbalance is particularly pronounced in rural areas, where aging rates often exceed 40% amid ongoing population decline, while healthcare resources remain concentrated in large metropolitan areas [1].

Hokkaido represents an instructive case, as its combination of

geographic dispersion, rapid population aging, and physician maldistribution mirrors challenges increasingly faced by rural regions in many high-income countries. Table 1 summarizes these disparities in Hokkaido. Sapporo, the capital city of Hokkaido, has more than 3.3 physicians per 1000 population. In contrast, rural areas such as Minami-Hiyama, where the aging rate exceeds 40%, have approximately 1.2 physicians per 1000 residents. According to a Japanese government report, the Physician Maldistribution Index in this region is 91.5—substantially lower than the national average (239)—indicating that the effective physician supply relative to healthcare demand is less than half of the national level [2,3]. Geographically, Esashi is located approximately 70 km (a 1.5-h drive) from Hakodate and about 300 km from Sapporo, with winter weather posing a significant barrier to access.

Traditionally, workforce policies have attempted to address these disparities through static physician allocation, such as providing

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financial incentives for rural practice. However, these approaches often treat rural regions as isolated resource-poor areas rather than integrated components of a broader health system. Institutional factors, such as limited schooling options for physicians' children, may further discourage long-term rural relocation for dual-career physician couples.

Relational ontology conceptualizes healthcare systems not as collections of independent entities, but as networks of relationships in which meaning and function emerge through interaction. Complementing this perspective, complex systems theory emphasizes interdependence and dynamic interactions within regional systems. Drawing on these perspectives, we developed a circulation-based model that links geographically distributed institutions through digital connectivity. Rather than merely redistributing physicians, this model enables the circulation of clinical experience, educational value, and professional development across regions, allowing both urban and rural systems to co-evolve.

In our design, physicians, trainees, patients, and clinical information circulate between urban and rural institutions through coordinated rotations, digital networks, and shared governance. Implemented since 2020 through collaboration between Sapporo Medical University and local governments, this study describes the design and early outcomes of this model.

## 2. Methods

### 2.1. Study setting and model structure

This study evaluates the implementation of a regional circulation-based healthcare model introduced in 2020, using a pre–post observational study to compare outcomes between the pre-implementation period (2015–2019) and the post-implementation period (2020–2023).

In Japan, healthcare delivery is organized through a hierarchical regional system consisting of primary, secondary, and tertiary (third) medical care areas. Secondary medical areas serve as the basic units for inpatient and emergency care planning, while tertiary centers provide advanced and specialized services.

Within this model, the Minami-Hiyama (including Esashi town)–Hakodate–Sapporo model links three functional tiers of healthcare delivery in southern Hokkaido (Table 1) [2–4]: a rural community within the Minami-Hiyama secondary medical area (centered on Esashi; population approximately 7000; physician density 1.2 per 1000 population), an intermediate regional hub within the Minami-Oshima secondary medical area (centered on Hakodate; population approximately 230,000; physician density 2.6 per 1000 population), and an urban tertiary center in the city of Sapporo (population approximately 1.9 million; physician density 3.3 per 1000 population in 2020) [2,3]. Geographically, Esashi is approximately 70 km (a 1.5-h drive) from Hakodate and approximately 280 km from Sapporo, with winter weather acting as a significant barrier.

Esashi Prefectural Hospital (EPH), the core hospital in this model, is

the only facility with a general ward and emergency department in Minami-Hiyama area. Thus, because nearly all healthcare interactions in the Minami-Hiyama area—including emergency admissions and outpatient visits—are concentrated at EPH, hospital data were considered representative of regional healthcare activity.

### 2.2. Physician rotation and training structure

The entire program was organized and supervised by a board-certified medical instructor. This senior physician provided both pedagogical oversight for the community based medical education (CBME) program and clinical governance for the rural–urban patient referral pathways. In practice, one or two trainees (medical students and/or residents) rotated between Sapporo and Esashi each month during a 24-week rotation period. In 2023, eight residents participated in the CBME program, each completing at least one month of training at EPH.

During this rotation, digital interactions—including 2–3 weekly web conferences per week—were conducted between supervising physicians at the urban site (Sapporo Medical University) and trainees and instructors at EPH.

### 2.3. Patient referral and care coordination

Patient care was organized through a tiered referral system. Patients requiring advanced medical care were transferred from EPH to hospitals in Hakodate or Sapporo, while relatively stable patients received ongoing care at EPH (Fig. 1).

### 2.4. Digital connectivity and system integration

High-resolution web-based teleconferencing systems and an interoperable electronic medical record (EMR) system supported continuity of care, with weekly case conferences linking Esashi and Sapporo.

For digital infrastructure, the ID-Link regional health information platform (NEC Nexsolutions Ltd., Japan) was utilized as a Fast Healthcare Interoperability Resources (FHIR)-compliant gateway to ensure interoperability among disparate electronic medical record (EMR) systems. Data exchange and encryption were performed in accordance with SS-MIX2 (Standardized Structured Medical Information eXchange) standards, enabling secure and standardized clinical data sharing across the participating regions. Remote access to electronic medical records during case conferences was conducted in compliance with the Japanese Ministry of Health, Labour and Welfare telemedicine guidelines to ensure the secure and appropriate handling of personal health information.

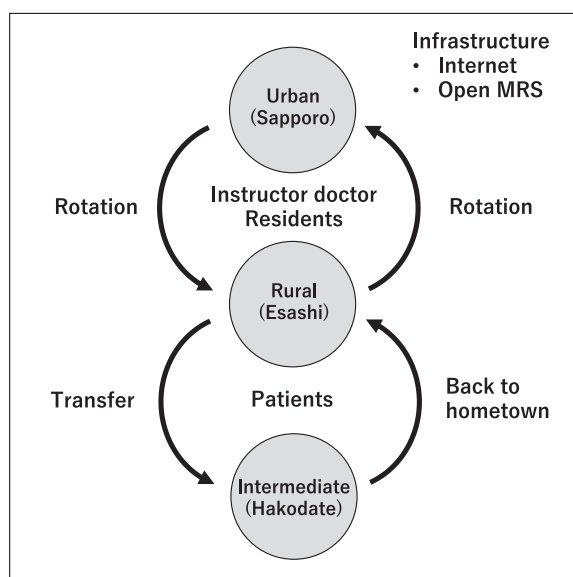
### 2.5. Ethics approval

This study does not involve individual-level patient data and thus did not require ethics committee approval.

**Table 1**  
Comparison of Medical Areas in Hokkaido and National Averages in Japan.

| Region                     | Major City | Population   | Area (km <sup>2</sup> ) | Aging Rate (≥65 years, %) | Physician Density (per 1000 population) | Physician Maldistribution |                                  |
|----------------------------|------------|--------------|-------------------------|---------------------------|-----------------------------------------|---------------------------|----------------------------------|
|                            |            |              |                         |                           |                                         | Index                     | National Rank (out of 335 areas) |
| Japan (national)           | –          | 125 million  | 377,975                 | 29                        | 2.5                                     | 239                       | –                                |
| Sapporo Medical Area       | Sapporo    | 2.36 million | 3547                    | 29                        | 3.3                                     | 297.2                     | 39th                             |
| Minami-Oshima Medical Area | Hakodate   | 470,000      | 3937                    | 36                        | 2.6                                     | 196.7                     | 106th                            |
| Minami-Hiyama Medical Area | Esashi     | 21,000       | 1423                    | 42                        | 1.2                                     | 91.5                      | 324th                            |

The Physician Maldistribution Index was calculated using the official methodology defined by the Ministry of Health, Labour and Welfare, Japan. This index is a government-reported indicator used in national health workforce policy and is calculated and published for each secondary medical area. National rankings are based on these official reports.



**Fig. 1.** Logic Model of the Circulation-Based Healthcare Framework; Three-layer rotation model linking urban (Sapporo), intermediate (Hakodate), and rural (Esashi) healthcare levels.

This schematic illustrates the circulation of key resources—physicians, medical students, and digital data—and its evaluated impact on regional health outcomes, medical education, and financial sustainability. In this model, the urban university in Sapporo provides human resources for medical education, including supervising physicians and trainees. Esashi Prefectural Hospital (EPH), located in a rural setting, functions both as a site of care and as a learning environment that enables work-based learning (WBL) through patient participation in medical education. Hakodate serves as an intermediate hub, delivering advanced medical care that cannot be sustained by EPH alone by accepting patients from Esashi, while also facilitating their return to their home community at Esashi after treatment. The Internet and Open Medical Record System (MRS), which supports patient circulation, functions as a foundational infrastructure that further promotes and sustains this circulation.

## 2.6. Statistical methods and definitions

All statistical analyses were performed using EZR (Saitama Medical Center, Jichi Medical University, version 1.61). Continuous variables are presented as mean (95% confidence interval [CI]). Comparisons between periods were conducted using Welch's *t*-test to account for unequal variances and small sample sizes, with  $p < 0.05$  considered statistically significant.

The standardized mortality ratio (SMR) is defined as the ratio of observed to expected deaths (with 100 indicating parity with the reference population) and was calculated for the Minami-Hiyama area using the 2015 Japanese national census as the reference [3, 5, 6]. SMRs were calculated for total mortality and for cause-specific categories (malignancy, cerebrovascular disease, heart disease, and pneumonia), stratified by sex.

**Financial data:** The operating ratio was defined as  $(\text{Total Operating Revenue} / \text{Total Operating Expenses}) \times 100$ . All monetary values are presented in Japanese Yen (JPY) without inflation adjustment, reflecting annual fiscal year reports. Operating ratio (%), outpatient revenue per patient (JPY), and inpatient revenue per patient (JPY) were analyzed.

**Educational outcomes** were described based on previously published cohort data and were not included in the statistical inference of the present study. The educational impact of the rotation program was evaluated using data from our prior cohort study of 97 medical students (Mihara et al., 2025 [8]), which was conducted with approval from the Institutional Review Board of Sapporo Medical University (Approval No. 5–1-15). In that study, students' self-confidence was assessed using a validated 5-point Likert scale across 15 domains.

## 3. Results

### 3.1. Clinical and institutional outcomes

Following implementation of the model in 2020, clinical outcomes in Minami-Hiyama showed overall improvement. The SMR demonstrated a downward trend in the post-implementation period. When comparing 2015–2019 with 2020–2023, the mean SMR decreased from 103.4 (95% CI: 98.2–108.6) to 98.3 (95% CI: 69.0–127.6) in men and from 104.8 (95% CI: 97.0–112.6) to 100.5 (95% CI: 87.3–113.7) in women, although these changes did not reach statistical significance (men:  $p = 0.63$ ; women:  $p = 0.47$ ). Disease-specific analyses revealed that improvements were more pronounced in acute conditions. In particular, pneumonia-related SMR showed significant reductions in both men ( $p = 0.01$ ) and women ( $p = 0.02$ ), while cerebrovascular disease demonstrated a borderline improvement in women ( $p = 0.06$ ). Changes in malignancy were modest.

Approximately 50 patients per year required inter-facility transfer during the study period, primarily for cardiovascular conditions. In FY2023, 52 patients were transferred from EPH to Hakodate for advanced care, with the majority transported by ambulance and a small number of critically ill patients transferred by helicopter to Sapporo Medical University. Despite this increased reliance on inter-facility transfer, mortality and adverse event rates among transferred patients were not higher than those observed in the general population of Hokkaido [6].

Institutional performance at EPH also improved during the study period. The operating balance ratio increased from 96.5% (95% CI: 95.4–97.6) in 2015–2019 to 106.9% (95% CI: 93.4–120.4) in 2020–2023, although this difference did not reach statistical significance ( $p = 0.07$ ). Outpatient revenue per patient increased from 11,218 JPY (95% CI: 11,209–11,227) to 11,605 JPY (95% CI: 11,163–12,047) ( $p = 0.09$ ), while inpatient revenue per patient increased significantly from 26,553 JPY (95% CI: 25,592–27,514) to 38,478 JPY (95% CI: 30,690–46,266) ( $p = 0.03$ ). In addition, total outpatient revenue rose from 514 million to 626 million yen, reflecting expanded clinical services and increased patient volume [7]. Rotational staffing also alleviated the workload of permanent clinicians, and participating physicians reported high levels of professional satisfaction.

### 3.2. Educational context

Educational findings from a previously reported study using the same cohort (Mihara et al., 2025 [8]) are summarized here for contextual reference. Participants reported improvements in confidence across domains such as emergency care, clinical skills, community medicine, and evidence-based medicine.

## 4. Discussion

The implementation of the Esashi–Hakodate–Sapporo circulation-based model was associated with improvements in regional health outcomes (SMR), institutional performance, and educational engagement, potentially contributing to workforce sustainability. By combining the rotation of physicians, residents, and patients with interoperable digital infrastructure, and linking a rural hospital, a regional referral hub, and an academic center, the system enables both clinical support and educational continuity. Through this circulation, rural hospitals are functionally integrated into the broader regional healthcare system, effectively sharing roles with urban institutions while maintaining local service capacity and supporting physician training (Table 2).

A key conceptual feature of this model is its focus on the productivity of resources used in workplace-based learning during medical education. In this context, clinical supervision is not separate from service delivery but inherently involves patient care, such that educational

**Table 2**  
Key Outcomes Before and After Implementation.

| Outcomes                             |                 |        | Average (95%CI)        |                        | p-value |
|--------------------------------------|-----------------|--------|------------------------|------------------------|---------|
|                                      |                 |        | Pre; 2015–2019         | Post; 2020–2023        |         |
| SMR                                  | Total           | Male   | 103.4                  | 98.3                   | 0.63    |
|                                      |                 | Female | 104.8                  | 100.5                  | 0.47    |
|                                      | Malignancy      | Male   | 112.0 (100.4–123.6)    | 95.0 (83.0–107.0)      | 0.16    |
|                                      |                 | Female | 105.0 (95.2–114.8)     | 108.8 (76.8–140.8)     | 0.81    |
|                                      | Cerebrovascular | Male   | 98.0 (86.1–109.9)      | 86.3 (47.7–124.9)      | 0.52    |
|                                      |                 | Female | 94.0 (86.2–101.8)      | 80.0 (71.0–89.0)       | 0.06    |
|                                      | Heart disease   | Male   | 121.0 (95.0–147.0)     | 90.0 (53.4–126.6)      | 0.22    |
|                                      |                 | Female | 110.0 (100.3–119.7)    | 105.0 (89.0–121.0)     | 0.63    |
|                                      | Pneumonia       | Male   | 65.0 (54.5–75.5)       | 35.0 (22.5–47.5)       | 0.01    |
|                                      |                 | Female | 77.0 (63.5–90.5)       | 43.8 (33.5–54.1)       | 0.02    |
| Operating balance ratio (%)          |                 |        | 96.5 (95.4–97.6)       | 106.9 (93.4–120.4)     | 0.07    |
| Outpatient revenue per patient (JPY) |                 |        | 11,218 (11,209–11,227) | 11,605 (11,163–12,047) | 0.09    |
| Inpatient revenue per patient (JPY)  |                 |        | 26,553 (25,592–27,514) | 38,478 (30,690–46,266) | 0.03    |

The standardized mortality ratio (SMR) was calculated for the Minami-Hiyama area using the 2015 Japanese national population as the reference. Values are presented as mean (95% confidence intervals). *P*-values were calculated using Welch's *t*-test. The study periods were defined as pre-implementation (2015–2019) and post-implementation (2020–2023). The following financial indicators were evaluated at Esashi Prefectural Hospital (EPH): operating balance ratio (%), outpatient revenue per patient (JPY), and inpatient revenue per patient (JPY).

activities simultaneously contribute to healthcare provision, professional development, and institutional performance. Rural healthcare settings, where trainees are directly engaged in clinical practice under supervision, function as highly productive environments in which care delivery and learning are co-produced.

While rotation-based systems are not new, this model reconceptualizes continuity by shifting its unit from the individual to the system. Rather than relying on prolonged individual placement—which may be constrained by career specialization pathways and family responsibilities in many contexts in Japan—it maintains clinical and educational continuity through coordinated circulation of physicians, trainees, patients, and clinical information. Supported by digital infrastructure and longer-term supervising physicians, this approach ensures sustained system-level continuity through repeated short-term rotations, allowing medical resources to remain continuously present in rural areas.

Currently, increasing specialization in large tertiary hospitals in Japan has made advanced clinical training more resource-intensive, while opportunities for basic clinical training may be relatively limited. In contrast, rural hospitals—supported by digital infrastructure and supervising physicians—offer highly productive environments for workplace-based learning, where clinical care and education are integrated. As a result, rural placements are not a trade-off in training quality but provide high-value opportunities for skill acquisition while contributing to healthcare delivery and system sustainability.

From an equity and global health perspective [9,10], this model aligns with the PROGRESS-Plus framework [11] and WHO Workforce 2030 [12], addressing urban–rural disparities through digitally enabled circulation of healthcare resources [13]. It also supports continuity of care for older rural populations by integrating digital consultations with existing primary care providers, and aligns with SDGs 3, 9, and 10. In Japan, the model is partly supported by the “hometown tax” system as a flexible local funding mechanism. However, given its minimum viable digital requirements (e.g., open-source EMRs and moderate bandwidth), the model may be adaptable to low- and middle-income countries (LMICs) [14], where alternative financing—such as international development funding or performance-based domestic mechanisms—may be required.

## 5. Limitations and Future Research

This study has several limitations. First, it was conducted in a single region, limiting generalizability. Second, the observational pre–post design without a control region limits causal inference, particularly given the overlap with the COVID-19 pandemic. Third, although we

used same cohort, educational outcomes were not analyzed with inferential statistics in this study and should be interpreted as contextual findings. Finally, long-term outcomes, including physician retention and workforce stability, were not assessed.

Future research should include multicenter implementation studies or interrupted time series analyses to better account for secular trends. In addition, patient-reported outcomes, long-term workforce retention, and formal cost-effectiveness analyses are needed to evaluate the sustainability and scalability of the model.

## 6. Conclusion

The circulation-based model implemented in southern Hokkaido demonstrates how physician mobility, digital connectivity, and regional collaboration can support the sustainability of rural healthcare systems. By linking rural hospitals with referral centers and academic institutions, the model expands the functional medical service area while maintaining clinical quality and educational continuity. Circulation-based system design may therefore provide a practical approach to addressing health workforce maldistribution in geographically dispersed regions.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT to improve the English language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

## CRedit authorship contribution statement

**Atsushi Jinno:** Project administration, Methodology, Conceptualization. **Yoshihisa Tsuji:** Writing – original draft, Supervision, Project administration, Methodology, Data curation, Conceptualization.

## Ethics approval

This study does not involve individual-level patient data and therefore did not require ethics committee approval. The educational evaluation component involving medical students has been previously published (Mihara H, et al. BMC Medical Education, 2025) and was conducted with approval from the Institutional Review Board of Sapporo Medical University (Approval No. 5–1-15). In the present study, we refer to this published work and include its findings as external evidence

to evaluate the outcomes of the proposed model.

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## Declaration of competing interest

The author declares no competing interests.

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